



Analysing solutions

Name:

Date:

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For which solution would it be scientifically correct to say that it contains 0.0050 mol of sodium hydroxide?

A 25.0 cm³ of 0.20 mol dm⁻³ B 10 cm³ of 0.5 mol dm⁻³

C 20.00 cm³ of 0.250 mol dm⁻³ D 2.5 cm³ of 0.50 mol dm⁻³

Correct Answer: A B C D

Explanation:

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A student has a 30 cm³ sample of dilute hydrochloric acid. The concentration of the acid is 1 mol dm⁻³. He wants the acid to have a concentration of 0.1 mol dm⁻³. How much water does he need to add to the 30 cm³ sample of acid?

A 30 cm³ B 270 cm³ C 300 cm³ D 330 cm³

Correct Answer: A B C D

Explanation:

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Calculate the volume of 1.0 mol dm⁻³ hydrochloric acid that can be neutralised by a milk of magnesia tablet containing 0.29 g of magnesium hydroxide.

$\text{Mg}(\text{OH})_2(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{MgCl}_2(\text{aq}) + 2\text{H}_2\text{O}(\text{l})$

A_r values: Mg = 24, O = 16, H = 1.

A 0.0050 cm³ B 0.010 cm³ C 5.0 cm³ D 10.0 cm³

Correct Answer: A B C D

Explanation:

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In a titration, a student uses sulfuric acid to neutralise sodium hydroxide.

$$\text{H}_2\text{SO}_4(\text{aq}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{Na}_2\text{SO}_4(\text{aq}) + 2\text{H}_2\text{O}(\text{l})$$
 He finds that 25.00 cm³ of 0.0880 mol dm⁻³ aqueous sodium hydroxide, NaOH, is neutralised by 17.60 cm³ of dilute sulfuric acid, H₂SO₄.
 What is the concentration of the sulfuric acid?

A 0.250 mol dm⁻³

B 0.125 mol dm⁻³

C 0.0625 mol dm⁻³

D 0.00125 mol dm⁻³

Correct Answer: A B C D

Explanation:

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